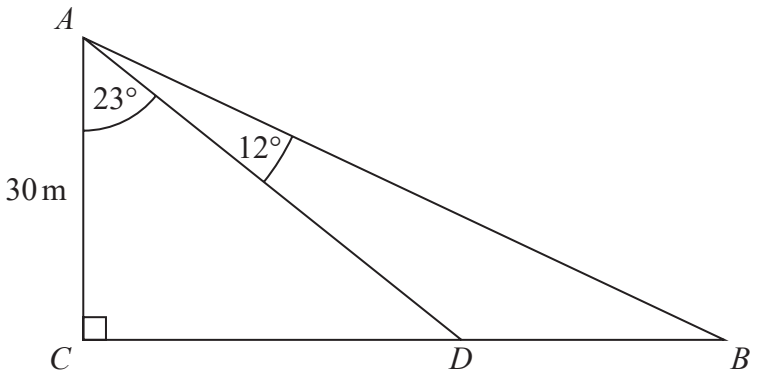
9 The diagram shows the right-angled triangle ABC.



CDB is a straight line.

Calculate the length of DB.
Give your answer correct to 2 decimal places.

Angle CAB = 23° + 12° = 35°.

In triangle ACD: tan 23° = CD ÷ 30

1 mark
Use opposite ÷ adjacent.

CD = 30 tan 23° = 12.73424449... m

1 mark
Find one base length.

In triangle ACB: tan 35° = CB ÷ 30

CB = 30 tan 35° = 21.00622615... m

DB = CB – CD = 8.27198165... m

1 mark
Subtract the two lengths.

DB = 8.27 m (2 decimal places)

1 mark
Round at the end.

..... m

(Total for Question 9 is 4 marks)

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